Derivatives: Contracts, Models, and Markets · Technical Note 38

Expected Shortfall as a Minimum, and Why It Is Coherent

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Chapter 38 of Derivatives: Contracts, Models, and Markets asserts that expected shortfall is a coherent risk measure, and the online note [TN41] *trains a hedging network on the representation of expected shortfall as a minimum over one auxiliary number, its equation (TN41.2). Neither result is proved in either place. This note proves both, and the first is a corollary of the second: once expected shortfall is written as the minimum of an expectation of a convex function, subadditivity takes one line. The derivation is done for an arbitrary loss distribution, atoms included, because Chapter 38's own counterexample has atoms and because the integral form of expected shortfall is the definition precisely where the distribution jumps. It then recovers the chapter's normal-book figures and its two-bond figures from the minimization. A further section derives the generalized Pareto results behind Chapter 38's tail estimation (the threshold limit for power and normal tails, the tail-quantile and expected-shortfall formulas, and the likelihood with its restriction and precision) and checks them against Example 38.7. Another section sets out why expected shortfall is hard to backtest and why it cannot be scored on its own, two points the chapter states without detail. A short final section works two calculations the chapter leaves out: the standard error of an expected-shortfall estimate, which follows from the minimization, and the Cornish–Fisher quantile of Section 38.7.*

`tools/verify/tn38-expected-shortfall.py` *recomputes every result here.*

1 THE TWO MEASURES FOR ANY DISTRIBUTION

Let L be the loss on a portfolio over the horizon, larger being worse, with distribution function $F(l) = \mathbb{P}(L \le l)$ and a finite mean. Chapter 38 writes the confidence level as c and [TN41], following the optimization literature, as α; they are the same number, and this note uses c throughout. [TN41] also calls expected shortfall by the optimization literature's name for it, conditional value at risk (CVaR).

Chapter 38 defines value at risk by $\mathbb{P}(L > \mathrm{VaR}_c) = 1 - c$, its equation (38.1). When F jumps over the level c that equation has no solution, and the definition that covers every case is the lower quantile, the generalized inverse of F:

$$\mathrm{VaR}_c = q(c) = \inf\{\, l : F(l) \ge c \,\}, \qquad 0 < c < 1. \tag{TN38.1}$$

It agrees with (38.1) whenever (38.1) has a solution, and where it has none it picks the loss at which F crosses c, which is how Example 38.2 reads its quantiles. The function q is nondecreasing, and because F is right-continuous it satisfies $q(u) \le l$ if and only if $u \le F(l)$.

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Expected shortfall is defined by Chapter 38's equation (38.2),

$$\mathrm{ES}_c = \frac{1}{1-c}\int_c^1 q(u)\,du, \tag{TN38.2}$$

the average of all the value-at-risk figures beyond c.

The integral can be read as an expectation. Let U be uniform on $(0,1)$. The equivalence just stated gives $\mathbb{P}\big(q(U) \le l\big) = \mathbb{P}\big(U \le F(l)\big) = F(l)$, so $q(U)$ has the distribution of L, and for any function h with a finite expectation

$$\mathbb{E}\big[h(L)\big] = \int_0^1 h\big(q(u)\big)\,du. \tag{TN38.3}$$

In this representation (TN38.2) is $\mathbb{E}\big[q(U)\,\mathbf{1}\{U > c\}\big]/(1-c)$: the mean loss over the worst $1-c$ of outcomes counted by probability. When F jumps over c, the uniform variable splits the atom at the quantile, keeping inside the tail exactly the part of it that the tail needs. The conditional mean $\mathbb{E}[L \mid L > \mathrm{VaR}_c]$ cannot do this, since it either takes the whole atom or none of it, and that is why the integral (38.2) and the conditional mean part company where the distribution jumps.

2 EXPECTED SHORTFALL AS A MINIMUM

For each real number w define

$$g(w) = w + \frac{1}{1-c}\,\mathbb{E}\big[(L-w)^+\big]. \tag{TN38.4}$$

The claim of equation (TN41.2) is that $\min_w g(w) = \mathrm{ES}_c$, attained at $w = \mathrm{VaR}_c$. The proof has four steps.

Convexity. For each loss l, the map $w \mapsto (l-w)^+$ is convex, and an expectation of convex functions is convex. Adding the linear term w preserves convexity, so g is convex; it is finite because L has a finite mean.

The slope. The right derivative of $(l-w)^+$ in w is $-\mathbf{1}\{l > w\}$ and the left derivative is $-\mathbf{1}\{l \ge w\}$. The difference quotients are bounded by one, so the derivative passes through the expectation, and

$$g'_+(w) = 1 - \frac{\mathbb{P}(L > w)}{1-c} = \frac{F(w)-c}{1-c}, \qquad g'_-(w) = \frac{F(w^-)-c}{1-c}, \tag{TN38.5}$$

where $F(w^-) = \mathbb{P}(L < w)$. A convex function is minimized at w exactly when its left slope there is nonpositive and its right slope nonnegative, which by (TN38.5) is the condition $F(w^-) \le c \le F(w)$.

The minimizer. The quantile $q(c)$ meets that condition: $F(q(c)) \ge c$ by right-continuity, and $F(l) < c$ for every $l < q(c)$ by the definition of the infimum, so $F(q(c)^-) \le c$. Value at risk therefore minimizes g. The set of minimizers is the closed interval from $q(c)$ to $\inf\{l : F(l) > c\}$, which is a single point unless F is flat at height c; on that interval g is constant, and any point of it would serve.

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The minimum. Since q is nondecreasing, $q(u) - q(c)$ is nonnegative for $u > c$ and nonpositive for $u \le c$. Hence $\big(q(u) - q(c)\big)^+ = \big(q(u) - q(c)\big)\,\mathbf{1}\{u > c\}$ for every u, not merely for almost every u, and (TN38.3) gives

$$\mathbb{E}\big[(L - q(c))^+\big] = \int_c^1 \big(q(u) - q(c)\big)\,du = \int_c^1 q(u)\,du - (1-c)\,q(c). \quad \text{(TN38.6)}$$

Nothing in this step assumed a density: it uses only that q is monotone, so it holds with atoms. Substituting into (TN38.4), the $q(c)$ terms cancel and

$$\min_{w\in\mathbb{R}} \left\{ w + \frac{1}{1-c}\,\mathbb{E}\big[(L-w)^+\big] \right\} = g\big(q(c)\big) = \frac{1}{1-c}\int_c^1 q(u)\,du = \mathrm{ES}_c, \quad \text{(TN38.7)}$$

which is equation (TN41.2), with the minimizing w the value at risk.

Evaluating g at $q = q(c)$ directly, without the quantile transform, gives the same number in a form that shows how the atom is handled:

$$\mathrm{ES}_c = \frac{1}{1-c}\Big(\mathbb{E}\big[L\,\mathbf{1}\{L > q\}\big] + q\,\big(F(q) - c\big)\Big). \quad \text{(TN38.8)}$$

The first term is the tail strictly beyond the quantile. The second is the share of the atom at the quantile that falls inside the worst $1 - c$, valued at the quantile. When F is continuous at q, $F(q) = c$, the second term vanishes and (TN38.8) is the conditional mean $\mathbb{E}[L \mid L > q]$, the form Chapter 38 gives for a continuous distribution.

3 WHAT TN41 USES

Three properties of (TN38.7) make it the form a training loop can use.

The first is that it removes the moving boundary. Written as $\mathbb{E}[L \mid L > \mathrm{VaR}_c]$, expected shortfall averages over a tail whose edge moves whenever the hedging parameters θ move. Written as (TN38.7), it is the minimum over (θ, w) jointly of the ordinary expectation of

$$G(\theta, w; L_\theta) = w + \frac{(L_\theta - w)^+}{1-c},$$

a piecewise-linear function of the loss with one kink. At fixed θ and w, a batch average of G estimates the objective without bias, which an average over the batch's own worst paths does not.

The second is that the gradient is carried by tail paths only. Path by path,

$$\frac{\partial G}{\partial w} = 1 - \frac{\mathbf{1}\{L_\theta > w\}}{1-c}, \qquad \frac{\partial G}{\partial \theta} = \frac{\mathbf{1}\{L_\theta > w\}}{1-c}\,\frac{\partial L_\theta}{\partial \theta}, \quad \text{(TN38.9)}$$

and the factor $\mathbf{1}\{L > w\}/(1-c)$ is the one [TN41] quotes. A path whose loss is below w contributes nothing to the gradient in θ; at $c = 0.95$ and a batch of 2,048 paths the expected number that contribute is $2{,}048 \times 0.05 = 102.4$, that note's "about a hundred." The expectation of the first expression is the right slope (TN38.5), so stochastic gradient descent in w pushes it toward the quantile of the current loss distribution. And at a minimizing w the slope in w is zero, so the derivative of ES_c in θ equals $\mathbb{E}\big[\mathbf{1}\{L_\theta > \mathrm{VaR}_c\}\,\partial L_\theta/\partial\theta\big]/(1-c)$: the movement of the tail's edge contributes nothing to first order, which is the difficulty the representation was introduced to remove.

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The third is that the kink is harmless. The function $(l - w)^+$ has no derivative at $l = w$, but when L_θ has a continuous distribution that event has probability zero, so (TN38.9) holds on almost every path. The integrand is Lipschitz in L_θ and w, so a bound on $\partial L_\theta / \partial \theta$ lets the derivative pass through the expectation by dominated convergence. This is the argument that licenses differentiating a call payoff pathwise at its strike.

Two practical consequences follow from the shape of g. Every w gives an upper bound, $g(w) \geq \mathrm{ES}_c$, so a w that has not yet converged overstates the risk and never understates it. And the bound is tight to second order: when L has a density f, $g(w) - \mathrm{ES}_c \approx f(q)\,(w - q)^2 / \big(2(1 - c)\big)$ near the quantile, so an error in w costs much less in the objective than in the quantile itself. Section 5 puts numbers on both.

4 COHERENCE

Chapter 38 calls a risk measure ρ coherent if it is monotonic, positively homogeneous, translation-invariant and subadditive. In this note's convention, where ρ is applied to losses X and Y, the four properties are:

- monotonicity: if $X \leq Y$ in every state, then $\rho(X) \leq \rho(Y)$;
- positive homogeneity: $\rho(\lambda X) = \lambda\, \rho(X)$ for every $\lambda \geq 0$;
- translation invariance: $\rho(X + m) = \rho(X) + m$ for every constant m, so adding a certain loss of m adds m to the measure;
- subadditivity: $\rho(X + Y) \leq \rho(X) + \rho(Y)$, Chapter 38's equation (38.3).

Each follows for ES_c from (TN38.7). Write g_X for the function (TN38.4) built on the loss X.

Monotonicity. If $X \leq Y$ then $(X - w)^+ \leq (Y - w)^+$, so $g_X(w) \leq g_Y(w)$ for every w, and the minimum of the smaller function is the smaller minimum.

Translation invariance. Substituting $v = w - m$,

$$g_{X+m}(w) = w + \frac{1}{1 - c}\,\mathbb{E}\big[(X - v)^+\big] = m + g_X(v),$$

and minimizing over w is minimizing over v, so $\mathrm{ES}_c(X + m) = \mathrm{ES}_c(X) + m$.

Positive homogeneity. For $\lambda > 0$, substituting $v = w/\lambda$,

$$g_{\lambda X}(w) = \lambda v + \frac{\lambda}{1 - c}\,\mathbb{E}\big[(X - v)^+\big] = \lambda\, g_X(v),$$

so $\mathrm{ES}_c(\lambda X) = \lambda\, \mathrm{ES}_c(X)$. At $\lambda = 0$ both sides are zero.

Subadditivity. Let w_X and w_Y minimize g_X and g_Y; the value-at-risk figures of X and Y will do. For any numbers a and b, $(a + b)^+ \leq a^+ + b^+$, so state by state

$$\big(X + Y - (w_X + w_Y)\big)^+ \leq (X - w_X)^+ + (Y - w_Y)^+.$$

Taking expectations, dividing by $1 - c$ and adding $w_X + w_Y$ to both sides gives $g_{X+Y}(w_X + w_Y) \leq g_X(w_X) + g_Y(w_Y)$, and the minimum of g_{X+Y} is no larger than its value at the particular point $w_X + w_Y$. So

$$\mathrm{ES}_c(X + Y) \leq g_{X+Y}(w_X + w_Y) \leq g_X(w_X) + g_Y(w_Y) = \mathrm{ES}_c(X) + \mathrm{ES}_c(Y). \qquad \blacksquare \tag{TN38.10}$$

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The argument requires nothing about the joint distribution of X and Y: they may be independent, dependent, or the same position twice. Nor does it require continuity, because (TN38.7) was proved with atoms; it holds for Example 38.2's bonds without modification.

Why the same argument fails for value at risk. Expected shortfall is the *value* of a minimization, and subadditivity came from an inequality between the functions being minimized. Value at risk is where the minimum is *attained*, and nothing in (TN38.10) says anything about where g_{X+Y} is smallest. In Example 38.2 the two bonds' minimizers are $w_X = w_Y = 0$, but the right slope of g_{X+Y} at $w_X + w_Y = 0$ is negative, so the pair's minimizer lies elsewhere, at \$10 million. In terms of quantiles, expected shortfall is an average of quantiles, and averaging over the tail is what produces the convexity in (TN38.10), while value at risk is a single quantile, and the quantile of a sum has no general relation to the sum of the quantiles. For jointly normal losses at any confidence above one half the relation happens to hold, which is why Chapter 38 finds value at risk well behaved on smooth, roughly elliptical distributions and unreliable on the books with jumps.

5 TWO CHECKS AGAINST CHAPTER 38

The normal book. Example 38.1 holds a position whose daily loss is normal with mean zero and standard deviation $\sigma = \$2.0$ million. For such a loss, with $k = w/\sigma$ and N and N' the standard normal distribution and density,

$$g(w) = w + \frac{\sigma N'(k) - w\left(1 - N(k)\right)}{1 - c}. \tag{TN38.11}$$

At $w = z\sigma$ with $z = N^{-1}(c)$ the term $w(1 - N(k))/(1 - c)$ equals w and cancels the leading term, leaving $g = \sigma N'(z)/(1 - c)$, the chapter's closed form for expected shortfall. Minimizing (TN38.11) numerically, by golden-section search with no knowledge of z, returns the same figures. The table below shows what the search sees at 99%:

w **(\$m)**	$g(w)$ **(\$m)**	$g(w) - \mathrm{ES}$	**Slope** $(F(w) - c)/(1 - c)$
4.00	5.70	0.37	−1.28
4.40	5.38	0.05	−0.39
4.65	5.33	0.00	0.00
4.90	5.37	0.04	0.29
5.50	5.68	0.35	0.70

The slope changes sign at the value at risk, and g lies above the expected shortfall everywhere else. Setting w about 5% below the true quantile, at \$4.40 million, overstates the expected shortfall by less than 1%, which is the second-order flatness of Section 3 in numbers.

The two bonds. Example 38.2's bonds are independent, \$10 million each, with a 0.8% default probability and zero recovery, measured at 99%. Each loss distribution is discrete, so g is piecewise linear with kinks at the atoms, and its minimum can be

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found exactly by evaluating it there. For one bond, on $0 \le w \le 10$,

$$g(w) = w + \frac{0.008 \times (10 - w)}{0.01} = 8 + 0.2\,w,$$

and for $w < 0$ the slope is -99, so the minimum is at $w = 0$, the bond's value at risk, and equals \$8 million, the chapter's figure. For the pair, the right slope at $w = 0$ is $(0.984064 - 0.99)/0.01 = -0.5936$ and at $w = 10$ it is $(0.999936 - 0.99)/0.01 = 0.9936$, so the minimum is at \$10 million and equals

$$g(10) = 10 + \frac{0.000064 \times 10}{0.01} = 10.064.$$

By (TN38.8) the same number is $\big(0.000064 \times 20 + 10 \times 0.009936\big)/0.01$: the second term is the share of the \$10 million atom that falls inside the worst 1%, which is the "remaining 0.009936 at \$10m" of the chapter's calculation.

Position	c	**Minimizer** w	**Minimum** g	**Chapter 38 prints**
Normal book	97.5%	3.9199	4.6756	VaR \$3.92m, ES \$4.68m
Normal book	99%	4.6527	5.3304	VaR \$4.65m, ES \$5.33m
One bond	99%	0	8.000	VaR \$0, ES \$8m
Two bonds	99%	10	10.064	VaR \$10m, ES \$10.06m

The subadditivity chain (TN38.10) reads $10.064 \le 16 \le 8 + 8$ on these bonds. The middle inequality is an equality here, because losses that are never negative satisfy $(a + b)^+ = a^+ + b^+$ at $w_X = w_Y = 0$; all of the diversification benefit comes from the first inequality, from moving w from 0 to 10. Value at risk, read off the minimizer column, goes from $0 + 0$ to 10.

Two further numbers show why the integral form must be the definition here. The conditional mean $\mathbb{E}[L \mid L > \mathrm{VaR}_{99\%}]$ is \$10 million for one bond and \$20 million for the pair, neither of which is the average loss over the worst 1%: it conditions on an event of probability 0.008 in the first case and 0.000064 in the second, not 0.01. And at $c = 0.992$, where $F(0) = c$ exactly, a single bond's g is constant at \$10 million for every w between 0 and 10: the flat interval of Section 2, on which any point is a value at risk and every one gives the same expected shortfall.

6 THE GENERALIZED PARETO TAIL

Section 38.9 estimates a 99.9% value at risk and expected shortfall from 500 days of losses by fitting the generalized Pareto law to the largest 25 of them, and Example 38.7 carries the calculation through. The chapter uses five results without deriving them: that the excesses over a high threshold follow that law, that a Student's t with ν degrees of freedom has $\xi = 1/\nu$ and the normal has $\xi = 0$, the formulas for the tail quantile and the expected shortfall, and the likelihood with its restriction $\xi \ge -1/2$. This section derives each of them. The exception is the general limit theorem, which is stated and then worked by hand for the two tails the chapter uses.

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The limit law. For a threshold u, the **excess distribution** is the distribution of the loss beyond u, given that u is exceeded:

$$F_u(y) = \mathbb{P}(L - u \le y \mid L > u) = \frac{F(u+y) - F(u)}{1 - F(u)}, \qquad y \ge 0. \tag{TN38.12}$$

The generalized Pareto distribution with shape ξ and scale $\psi > 0$ is

$$G_{\xi,\psi}(y) = 1 - \left(1 + \frac{\xi\, y}{\psi}\right)^{-1/\xi}, \qquad G_{0,\psi}(y) = 1 - e^{-y/\psi}, \tag{TN38.13}$$

on $y \ge 0$ when $\xi \ge 0$ and on $0 \le y \le -\psi/\xi$ when $\xi < 0$; the exponential is the limit of the first form as $\xi \to 0$. The Pickands–Balkema–de Haan theorem says that for every distribution in the domain of attraction of an extreme value law, which includes every continuous distribution used for losses in practice, there is a scale function $\psi(u)$ with

$$\sup_y \left| F_u(y) - G_{\xi,\psi(u)}(y) \right| \to 0$$

as u rises to the right end of the distribution, with the shape ξ fixed by F. Its proof is beyond this note. The two cases the chapter uses can be done directly, and they show where the scale function comes from.

A power tail has $\xi = 1/\alpha$. Suppose $\mathbb{P}(L > x) = Cx^{-\alpha}$ for all large x. Then for u in that range

$$\mathbb{P}(L - u > y \mid L > u) = \left(\frac{u+y}{u}\right)^{-\alpha} = \left(1 + \frac{\xi\, y}{\xi u}\right)^{-1/\xi}, \qquad \xi = \frac{1}{\alpha},$$

which is $G_{\xi,\,\xi u}$ exactly. A tail that is a power only asymptotically, $\mathbb{P}(L > x) = x^{-\alpha}\mathcal{L}(x)$ with $\mathcal{L}(tx)/\mathcal{L}(x) \to 1$, picks up a factor $\mathcal{L}(u+y)/\mathcal{L}(u)$ that tends to one, and the limit is the same. A Student's t with ν degrees of freedom has density proportional to $(1 + t^2/\nu)^{-(\nu+1)/2}$, which falls like $t^{-(\nu+1)}$; its tail therefore falls like $t^{-\nu}$, and $\xi = 1/\nu$. For $\nu = 4$ the distribution function has the closed form $F(t) = 1/2 + 3(\kappa - \kappa^3/3)/4$ with $\kappa = t/\sqrt{t^2+4}$, and expanding in $1 - \kappa \approx 2/t^2$ gives

$$\mathbb{P}(T > t) \approx 3\, t^{-4}, \tag{TN38.14}$$

a power tail with $\alpha = 4$, which is the chapter's $\xi = 0.25$. Rescaling a loss multiplies the tail's constant and leaves its exponent alone, so the book's scaled t has the same shape.

A normal tail has $\xi = 0$. For the standard normal, $1 - N(x) = N'(x)/x \cdot \big(1 + O(x^{-2})\big)$, so

$$\frac{1 - N(u+y)}{1 - N(u)} \approx \frac{u}{u+y}\, e^{-uy - y^2/2}.$$

Put $y = \tau/u$ and this is $(1 + \tau/u^2)^{-1} e^{-\tau - \tau^2/(2u^2)} \to e^{-\tau}$. The excess, measured in units of $1/u$, is exponential: $\xi = 0$, with $\psi(u) = 1/u$, or σ^2/u for a normal loss with mean zero and standard deviation σ. The scale shrinks as the threshold rises, so the further out one looks, the more tightly the remaining losses bunch just above the threshold. That is what a tail that dies off faster than any power amounts to, and the lognormal and the gamma land at $\xi = 0$ for the same reason.

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The limit arrives slowly. At the 95% threshold of Example 38.7's book, (TN38.14) is still far from the truth. The standardized threshold is $t = 2.1318$, a loss of $u = 2.1318\sqrt{2} = \$3.015$ million, where $3t^{-4} = 0.145$ against a true tail probability of 0.05. Taking the limit at its word, $\xi = 0.25$ and $\psi = \xi u = 0.754$, puts the 99.9% value at risk at \$8.02 million rather than \$10.14 million. A fit to the excesses does not recover those parameters either. The generalized Pareto law that best fits the exact excess distribution above that threshold, in the sense of maximizing the expected log-likelihood, which is what maximum likelihood converges to as the number of exceedances grows, has $\xi^* = 0.192$ and $\psi^* = 1.222$. It puts the value at risk at \$10.13 million and the expected shortfall at \$13.33 million, against the true \$10.14 million and \$13.70 million. The method works at this threshold not because the limit has arrived but because the fitted shape and scale trade against each other, ξ low and ψ high, to reproduce the quantile. That is why peaks over threshold fits ψ freely rather than setting it to ξu, and why at a finite threshold $\hat{\xi}$ estimates the shape that fits the shoulder of the tail, which need not be the tail index.

Threshold stability. If Y has the law $G_{\xi,\psi}$ and $v \geq 0$ lies in its support, then

$$\mathbb{P}(Y - v > y \mid Y > v) = \left(\frac{1 + \xi(v + y)/\psi}{1 + \xi v/\psi}\right)^{-1/\xi} = \left(1 + \frac{\xi\, y}{\psi + \xi v}\right)^{-1/\xi}, \quad \text{(TN38.15)}$$

so the excess over a higher threshold is again generalized Pareto, with the same shape and the scale $\psi + \xi v$. The mean of $G_{\xi,\psi}$ is the integral of its tail, $\int_0^\infty (1 + \xi y/\psi)^{-1/\xi}\, dy = \psi/(1 - \xi)$, finite only for $\xi < 1$; in the same way the kth moment is finite only for $\xi < 1/k$, which matches a Student's t having moments of order below ν and no others. Together the two facts give a mean excess that is linear in the level:

$$\mathbb{E}[L - x \mid L > x] = \frac{\psi + \xi(x - u)}{1 - \xi}, \qquad x \geq u. \quad \text{(TN38.16)}$$

The tail quantile. For $x \geq u$, $\mathbb{P}(L > x) = \mathbb{P}(L > u)\,\mathbb{P}(L - u > x - u \mid L > u)$. Estimating the first factor by the fraction N_u/n of the n observations that exceed u, and the second by the fitted law,

$$\hat{\mathbb{P}}(L > x) = \frac{N_u}{n}\left(1 + \frac{\xi(x - u)}{\psi}\right)^{-1/\xi}. \quad \text{(TN38.17)}$$

Setting this equal to $1 - c$ and solving for x,

$$\mathrm{VaR}_c = u + \frac{\psi}{\xi}\left[\left(\frac{n}{N_u}(1 - c)\right)^{-\xi} - 1\right], \quad \text{(TN38.18)}$$

which tends to $u - \psi \ln\big((n/N_u)(1 - c)\big)$ as $\xi \to 0$. It applies only to $1 - c \leq N_u/n$, a quantile beyond the threshold. In Example 38.7, $(n/N_u)(1 - c) = 20 \times 0.001 = 0.02$: the 99.9% quantile of the loss is the point beyond which 2% of the excesses lie.

The expected shortfall. The fitted distribution is continuous, so by (TN38.8) the expected shortfall is the conditional mean beyond the value at risk, and (TN38.16) at $x = \mathrm{VaR}_c$ gives

$$\mathrm{ES}_c = \mathrm{VaR}_c + \frac{\psi + \xi(\mathrm{VaR}_c - u)}{1 - \xi} = \frac{\mathrm{VaR}_c + \psi - \xi u}{1 - \xi}, \quad \text{(TN38.19)}$$

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the chapter's formula. It is finite only for $\xi < 1$, and because it divides by $1 - \xi$, an estimated shape near one inflates it without bound. That is why, in the simulation after Example 38.7, the mean expected shortfall over the 400 windows sits above the median window's. As $c \to 1$ the value at risk grows without bound while ψ and u stay fixed, so

$$\frac{\mathrm{ES}_c}{\mathrm{VaR}_c} \to \frac{1}{1-\xi}.$$

For $\xi = 0.25$ the limit is $4/3$, against the exact ratio of 1.350 for the chapter's book at 99.9%. For a normal tail the ratio tends to one. A thin tail's two measures converge as the confidence level rises, while a fat tail keeps them a fixed proportion apart, which is the pattern in Example 38.6's table.

The likelihood. Differentiating (TN38.13) gives the density $(1/\psi)(1 + \xi y/\psi)^{-1/\xi - 1}$, and the log-likelihood of excesses $y_1, \dots, y_N$ (here N is the number of exceedances, N_u above) is

$$\ell(\psi, \xi) = -N \ln \psi - \left(1 + \frac{1}{\xi}\right) \sum_{i=1}^{N} \ln\!\left(1 + \frac{\xi\, y_i}{\psi}\right), \tag{TN38.20}$$

subject to $1 + \xi y_i/\psi > 0$ for every i. That constraint binds only when $\xi < 0$, where it says the largest excess lies below the endpoint $-\psi/\xi$, and it is what rules out $\xi < -1$. Let ψ fall toward $-\xi\, y_{\max}$, bringing the endpoint down onto the largest observation. That observation's term $\ln(1 + \xi y_{\max}/\psi)$ goes to $-\infty$, and its coefficient $-(1 + 1/\xi)$ is negative when $\xi < -1$, so the product goes to $+\infty$, $\ell \to +\infty$, and the likelihood has no maximum. (Equivalently, the density exponent $-1/\xi - 1$ is negative, so the fitted density is unbounded at the endpoint.) For $-1 < \xi < -1/2$ a maximum exists but the estimator lacks the usual normal approximation, which is why the stronger restriction $\xi \geq -1/2$ is imposed; for $\xi > -1/2$ it has it. Restricting to $\xi \geq -1/2$ costs nothing when the truth is a fat tail, and in the 400 simulated windows of Section 38.9 it binds in 14.

One dimension instead of two. Write $\vartheta = \xi/\psi$. Then $\ln(1 + \xi y_i/\psi) = \ln(1 + \vartheta y_i)$, and with $\psi = \xi/\vartheta$,

$$\ell = -N \ln \xi + N \ln \vartheta - \left(1 + \frac{1}{\xi}\right) S(\vartheta), \qquad S(\vartheta) = \sum_{i=1}^{N} \ln(1 + \vartheta y_i).$$

For fixed ϑ this is maximized over ξ where $-N/\xi + S/\xi^2 = 0$, that is, at $\xi(\vartheta) = S(\vartheta)/N$, and substituting back leaves a function of one variable,

$$\ell_p(\vartheta) = -N \ln \frac{\xi(\vartheta)}{\vartheta} - N\, \xi(\vartheta) - N, \qquad \ell_p(0) = -N \ln \bar{y} - N, \tag{TN38.21}$$

the second form being the exponential case. Since $\xi'(\vartheta) = \sum_i y_i/(1 + \vartheta y_i)/N > 0$, the restriction $\xi \geq -1/2$ is a lower bound on ϑ, and the support constraint is $\vartheta > -1/y_{\max}$. The verification script maximizes (TN38.21) by a grid followed by golden-section search in all 400 simulated windows of Section 38.9; in window 1 it returns $\hat{\vartheta} = 0.0854$, hence $\hat{\xi} = 0.0983$ and $\hat{\psi} = 1.1511$.

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How precise the shape is. For $\xi > -1/2$ the estimates are approximately normal in large samples, with covariance the inverse of the Fisher information:

$$\operatorname{Cov}\begin{pmatrix}\hat{\psi}\\ \hat{\xi}\end{pmatrix} \approx \frac{1+\xi}{N}\begin{pmatrix}2\psi^2 & -\psi\\ -\psi & 1+\xi\end{pmatrix}. \tag{TN38.22}$$

Two consequences bear on Example 38.7 and the simulation after it. The standard error of $\hat{\xi}$ is $(1+\xi)/\sqrt{N}$, which at $N = 25$ and $\xi = 0.25$ is 0.25, as large as the shape being estimated. Across the 400 windows the standard deviation of $\hat{\xi}$ is 0.28 and its median 0.12, so window 1's 0.098 is an ordinary draw rather than an unlucky one. And the correlation between the two estimates is $-1/\sqrt{2(1+\xi)} = -0.63$: a window that understates the shape tends to overstate the scale, and the two errors partly cancel in the quantile, as they did in the large-sample fit above. The windows show the cancellation at work. Their mean $\hat{\xi}$ is 0.107, below both the tail index and the 0.192 an unlimited sample would converge to, yet their mean value-at-risk estimate is within 0.3% of the truth.

7 BACKTESTING EXPECTED SHORTFALL, AND ELICITABILITY

Chapter 38 says two things about checking expected shortfall against outcomes: that it cannot be backtested by counting exceptions, and that it is not elicitable on its own while the pair of value at risk and expected shortfall is. They are separate problems. The first is about testing one model; the second is about ranking several.

Testing one model. A value-at-risk forecast predicts an event, $L_t > \mathrm{VaR}_t$, and a count of those events is binomial under the model. An expected-shortfall forecast predicts a conditional mean, so a test must compare tail losses with the forecast rather than count them. Acerbi and Székely (2014) give two statistics. Write $I_t = \mathbf{1}\{L_t > \mathrm{VaR}_t\}$ for an exception on day t of T, and $N_T = \sum_t I_t$. Then

$$Z_1 = \frac{1}{N_T}\sum_{t=1}^{T}\frac{L_t\, I_t}{\mathrm{ES}_t} - 1, \qquad Z_2 = \sum_{t=1}^{T}\frac{L_t\, I_t}{T\,(1-c)\,\mathrm{ES}_t} - 1. \tag{TN38.23}$$

If the forecasts are right, $\mathbb{E}[L_t \mid L_t > \mathrm{VaR}_t] = \mathrm{ES}_t$, so Z_1 averages ratios whose expectation is one and is centered on zero. It looks only at the size of the losses on exception days, not at how many there were, which is why it is run beside an exception count. Z_2 uses $\mathbb{E}[L_t I_t] = (1-c)\,\mathrm{ES}_t$, which follows from (TN38.8), so it too is centered on zero and responds both to too many exceptions and to exceptions that are too large. Positive values in either mean the forecast understated the tail. Neither statistic has a standard distribution, so its significance is found by simulating T days from the model's own forecast distributions and reading off where the observed value falls.

Ranking models. A statistic $T(F)$ of a distribution is **elicitable** if some scoring function $S(x, y)$, comparing a forecast x with an outcome y, has an expected score $\mathbb{E}_F[S(x, Y)]$ minimized at $x = T(F)$ for every F in the class. Competing forecasts of an elicitable statistic can then be ranked by their average score over a history, and the forecaster has no incentive to report anything but the truth. The mean is elicitable with

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$S = (x - y)^2$. A quantile is elicitable with the pinball score

$$S(x, y) = \big(\mathbf{1}\{y \le x\} - c\big)(x - y), \tag{TN38.24}$$

whose expectation has derivative $F(x) - c$ in x, zero at the c-quantile, so value at risk is elicitable. Expected shortfall is not. Gneiting (2011) observed that any elicitable statistic has convex level sets: if two distributions share its value, so does every mixture of them, because the mixture's expected score is the same mixture of two functions minimized at the same point. Expected shortfall fails this. Two distributions can have the same expected shortfall with different value-at-risk figures, and a mixture of them then has a quantile in a different place and an expected shortfall that is not the common value.

The pair. Fissler and Ziegel (2016) showed that value at risk and expected shortfall together are elicitable. One member of their family, written for losses and positive forecasts v and e, is

$$S(v, e; y) = \frac{\mathbf{1}\{y > v\}\,(y - v)}{(1 - c)\,e} + \frac{v}{e} + \ln e - 1. \tag{TN38.25}$$

Its expectation is minimized where the derivative in v, $\big(1 - \mathbb{P}(L > v)/(1 - c)\big)/e$, vanishes, which is at $v = q(c)$; and where the derivative in e vanishes, which is at

$$e = v + \frac{1}{1 - c}\,\mathbb{E}\big[(L - v)^+\big],$$

the expression minimized in (TN38.7), equal to ES_c at $v = q(c)$. The minimization that makes expected shortfall coherent is therefore also what makes it scorable, once the quantile is reported alongside it. For a standard normal loss at $c = 97.5\%$ the expected score (TN38.25) is smallest at $(v, e) = (1.9600,\ 2.3378)$, the two figures of Chapter 38's Example 38.1, which the verification script confirms by direct search.

What the framework does with this. The capital framework of Chapter 39 sets capital on expected shortfall and backtests value at risk, at 97.5% and 99%, by counting. That checks the quantile, which is elicitable and testable by count, and leaves the size of the tail to the attribution test and to supervisory judgment; Z_1 and Z_2 are run by banks for their own purposes but are not part of the rule.

8 TWO CALCULATIONS FROM CHAPTER 38

Chapter 38 states two figures whose working it leaves out for space: the precision of a 97.5% expected shortfall estimated from 500 days, in Section 38.3, and the Cornish–Fisher quantile of Section 38.7.

The standard error of expected shortfall. From n losses, the estimator is the sample form of (TN38.7),

$$\widehat{\mathrm{ES}}_c = \hat{q} + \frac{1}{n(1 - c)} \sum_{i=1}^{n} (L_i - \hat{q})^+,$$

with $\hat{q}$ the sample quantile. Because g is flat at its minimum (Section 3), the error in $\hat{q}$ moves the estimate only to second order, so to first order its variance is that of the

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sample mean of $(L-q)^+$, divided by $(1-c)^2$. For a continuous loss $\mathbb{P}(L>q) = 1-c$, and (TN38.6) gives the first moment, so

$$\begin{aligned}
\mathbb{E}\big[(L-q)^+\big] &= (1-c)\,(\mathrm{ES}_c - q),\\
\mathbb{E}\big[((L-q)^+)^2\big] &= (1-c)\,\mathbb{E}\big[(L-q)^2 \mid L>q\big] = (1-c)\big[\mathrm{Var}(L \mid L>q) + (\mathrm{ES}_c - q)^2\big],\\
\mathrm{Var}\big((L-q)^+\big) &= (1-c)\big[\mathrm{Var}(L \mid L>q) + (\mathrm{ES}_c - q)^2\big] - (1-c)^2(\mathrm{ES}_c - q)^2\\
&= (1-c)\big[\mathrm{Var}(L \mid L>q) + c\,(\mathrm{ES}_c - q)^2\big],
\end{aligned}$$

and dividing by $n(1-c)^2$ gives

$$\mathrm{s.e.}\big(\widehat{\mathrm{ES}}_c\big) \approx \sqrt{\frac{\mathrm{Var}(L \mid L>\mathrm{VaR}_c) + c\,(\mathrm{ES}_c - \mathrm{VaR}_c)^2}{n\,(1-c)}}. \qquad \text{(TN38.26)}$$

The first term is the spread of the losses inside the tail; the second is the uncertainty in how many losses fall there. Example 38.3's book has a daily loss of 88.39 thousand dollars times a standard t_4 variable. At $c = 97.5\%$ its value at risk is $88.39 \times 2.7764 = \$245.4$ thousand. The tail moments follow from the t_4 density in closed form: with $t = 2.7764$ and $\kappa = t/\sqrt{t^2+4}$, a standard t_4 variable T has

$$\mathbb{E}[T \mid T>t] = \frac{4}{(t^2+4)^{3/2}\,(1-c)} = 3.9936, \qquad \mathbb{E}[T^2 \mid T>t] = \frac{1-\kappa^3}{1-c} = 18.632,$$

so the expected shortfall is $88.39 \times 3.9936 = \$353.0$ thousand and the standard deviation of the losses beyond the quantile is $88.39\sqrt{18.632 - 3.9936^2} = \$144.8$ thousand. With $n = 500$,

$$\mathrm{s.e.}\big(\widehat{\mathrm{ES}}_{97.5\%}\big) = \sqrt{\frac{144.8^2 + 0.975 \times (353.0 - 245.4)^2}{500 \times 0.025}} = \$50.8 \text{ thousand},$$

14.4% of the expected shortfall. Section 38.5 finds \$45 thousand on the book's \$331 thousand 99% VaR, also 14%: the move from one measure to the other made the requirement sensitive to the shape of the tail without making it more precise.

The Cornish–Fisher quantile of the sold puts. Section 38.7's delta–gamma loss has $\mu = 0.020$, $\sigma = 0.2020$ (\$m), skewness $\gamma_1 = 0.590$ and excess kurtosis $\gamma_2 = 0.466$. At 99%, $z = 2.3263$, and the expansion is

$$\begin{aligned}
z_{\mathrm{CF}} = 2.3263 &+ \frac{2.3263^2 - 1}{6} \times 0.590 + \frac{2.3263^3 - 3 \times 2.3263}{24} \times 0.466\\
&- \frac{2 \times 2.3263^3 - 5 \times 2.3263}{36} \times 0.590^2 = 2.738.
\end{aligned}$$

The three corrections are 0.434, 0.109 and −0.131. The normal quantile alone gives $0.020 + 2.3263 \times 0.2020 = \$0.490$m; the skewness term alone, $0.020 + 2.760 \times 0.2020 = \$0.578$m; all three, $0.020 + 2.738 \times 0.2020 = \$0.573$m, against the exact \$0.5735m. The two fourth-order terms nearly cancel, so the skewness term does almost all the work, and it overshoots by about \$4,000.

FURTHER READING

Artzner, Delbaen, Eber and Heath (1999) is the paper that stated the four axioms, named the measures that satisfy them coherent, and showed that value at risk fails subadditivity.

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Rockafellar and Uryasev (2000) introduced the minimization (TN38.7) for continuous distributions as a way to optimize portfolios by linear programming, and their 2002 paper extends it to general distributions, which is the case this note treats. Acerbi and Tasche (2002) is short and settles what happens with atoms: it shows that the integral form (TN38.2), and equivalently (TN38.8), is coherent for every distribution, while the conditional mean is not, and it is the place to read the distinction made in Section 1. The threshold limit of Section 6 is due to Balkema and de Haan (1974) and Pickands (1975). Smith (1985) settles when the likelihood is regular, which is the source of the restriction $\xi \geq -1/2$, and Smith (1987) gives the covariance (TN38.22). Grimshaw (1993) is the reduction to one dimension. McNeil and Frey (2000) apply peaks over threshold to financial losses, and Embrechts, Klüppelberg and Mikosch (1997) is the standard reference for the theory as a whole. For Section 7, Acerbi and Székely (2014) is the source of the two test statistics, Gneiting (2011) of the elicitability argument, and Fissler and Ziegel (2016) of the joint scoring functions.

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